

NETWORK DESIGN

& Configuration Plan

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Environment: Small Enterprise Network

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Tool: Cisco Packet Tracer

Hosts	50 Devices across 4 VLANs
Topology	Three-tier hierarchical (Core / Distribution / Access)
Routing	OSPF with static default route to ISP
Security	ACLs, NAT, Port Security, DHCP Snooping

Network Overview

This document outlines the design and configuration of a secure, scalable small enterprise network. The three-tier hierarchical model separates core switching, distribution, and access layers to maximize performance, redundancy, and manageability.

VLAN DESIGN

VLAN Segmentation

VLAN ID	Name	Subnet	Gateway	Purpose
10	MANAGEMENT	192.168.10.0/24	192.168.10.1	Network devices & admin
20	STAFF	192.168.20.0/24	192.168.20.1	Employee workstations
30	SERVERS	192.168.30.0/24	192.168.30.1	Internal servers
40	GUEST	192.168.40.0/24	192.168.40.1	Guest WiFi (isolated)

IP ADDRESSING

IP Address Plan

Device	Interface	IP Address	Role
Core-SW-01	Vlan10	192.168.10.1	Default gateway MGMT
Core-SW-01	Vlan20	192.168.20.1	Default gateway STAFF
Core-SW-01	Vlan30	192.168.30.1	Default gateway SERVERS
Core-SW-01	Vlan40	192.168.40.1	Default gateway GUEST
Router-01	Gi0/0	10.0.0.1/30	WAN uplink to ISP
Router-01	Gi0/1	192.168.10.2	LAN interface
DNS-Server	eth0	192.168.30.10	Internal DNS
DHCP-Server	eth0	192.168.30.11	DHCP for all VLANs

Security Controls

Port Security

Enabled on all access layer switchports. Maximum 2 MAC addresses per port. Violation action: shutdown.

DHCP Snooping

Enabled on all VLANs. Trusted ports limited to uplinks and DHCP server port only.

ACL — Guest Isolation

Access list denies Guest VLAN (40) from reaching Staff (20) and Servers (30) subnets.

NAT Overload (PAT)

All internal traffic NATed through single public IP on Router-01 Gi0/0 interface.

OSPF Authentication

MD5 authentication enabled on all OSPF-participating interfaces to prevent route injection.

TESTING & VERIFICATION

Test Results

Test Case	Expected	Result
VLAN 20 ping to VLAN 30	Success via L3 switch	PASS
VLAN 40 ping to VLAN 20	Blocked by ACL	PASS
VLAN 20 internet access	Success via NAT	PASS
DHCP lease — VLAN 20	IP from 192.168.20.0/24	PASS
Port security violation	Port shuts down on 3rd MAC	PASS
OSPF neighbour formation	Full adjacency established	PASS